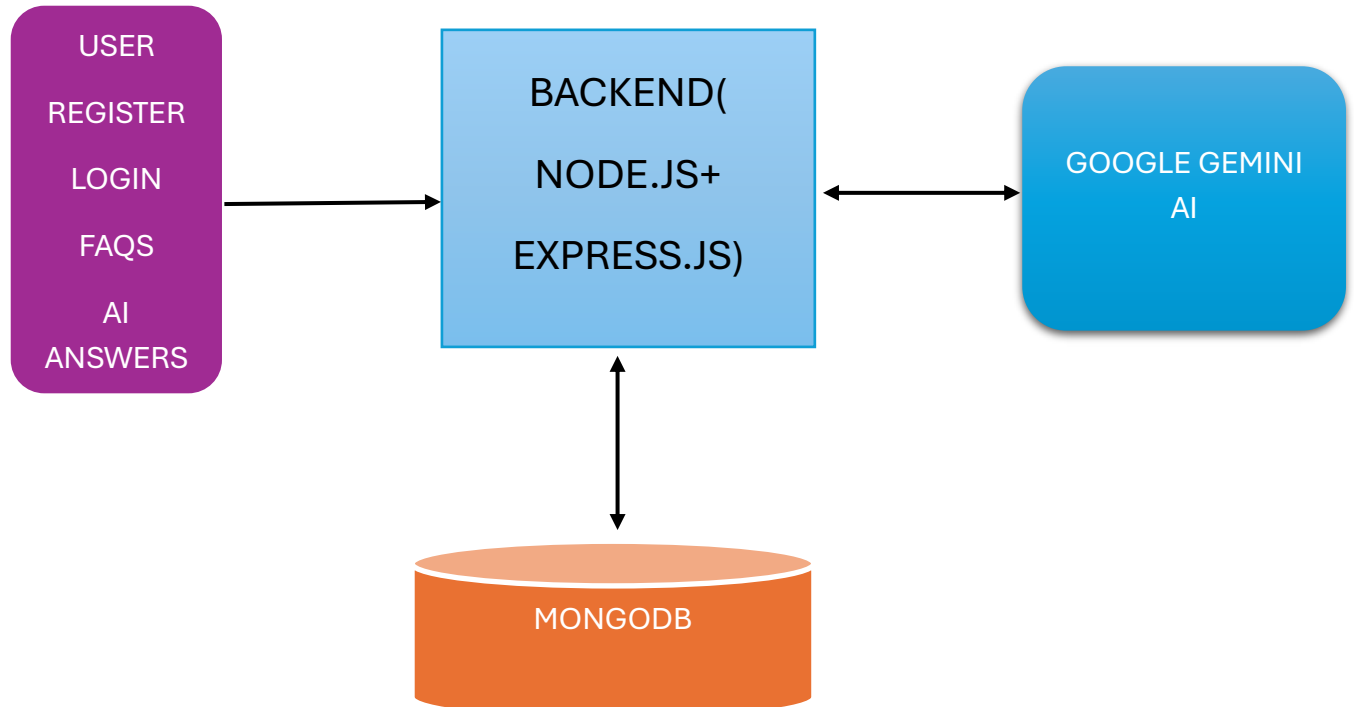


Project Design Phase

Solution Architecture Template

DATE:	24/09/2026
TEAM ID:	SWTID-2026-6715
PROJECT NAME:	AI FAQ Assistant API
MAXIMUM MARKS:	2Marks

Solution Architecture:



TECHNOLOGY STACK:

- Node.js
- Express.js
- MongoDB
- JWT
- Bcrypt
- Gemini AI

Explanation:

1. User Layer - Users interact with the system through a web interface to register, log in,

manage FAQs and search for answers.

2. Backend - Node.js and Express.js handle authentication (JWT + bcrypt), FAQ management, semantic search and AI integration.

3. AI Service - Google Gemini API generates answers, suggests topics and understands context.

4. Database MongoDB stores user data, FAQ records and search information.

5. Communication - All components communicate through secure REST APIs over HTTPS